

Course syllabus for

Academic year 2025/2026 >

SEE145 - Exoplanets and the solar system

Exoplaneter och solsystemet

Course syllabus adopted 2023-02-07 by Head of Programme (or corresponding)

Owner: [MPPHS](#)

7,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: Second-cycle

Main field of study: Engineering Physics

Department: 70 - SPACE, EARTH AND ENVIRONMENT

Teaching language: English

Application code: 85142

Open for exchange students: Yes

Block schedule: [A](#)

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 
0122 Examination 7,5 c Grading: TH		7,5 c					Mon 12/01-2026 pm J, Fri 10/04-2026 pm J, Wed 19/08-2026 pm J

In programs

[MPPHS Physics, Year 1 \(compulsory elective\)](#)

[MPPHS Physics, Year 2 \(elective\)](#)

Examiner:

[Carina Persson](#)



[Go to course homepage](#)

Eligibility

General entry requirements for master's level (second cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

English 6 (or by other approved means with the equivalent proficiency level)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Basic undergraduate physics (including mechanics, electrodynamics, quantum physics) and mathematics.

Aim

This course provides an introduction to our own solar system and planets orbiting other stars - exoplanets. Thousands of exoplanets have been discovered in recent decades following the development of groundbreaking instruments and facilities, particularly after the launch of several dedicated space telescopes. One of the greatest discoveries is the exceptional diversity of exoplanets where several new types of planets without equivalents in our own solar system have been discovered. The aim is to make students familiar with this new research field and gain an understanding of our own planet and planetary system in this context.

Learning outcomes (after completion of the course the student should be able to)

- Describe the structure of the solar system and the fundamental properties and physical processes of the planets and moons, including calculations of the planets' orbits.
- Have an understanding of different detection methods of exoplanets, including the possibilities and limitations of each method and basic application to observational data.
- Give an overview of planet formation theories and how these can explain the diversity of exoplanets and the architecture of planetary systems.

Content

- Introduction to stars.
- Structure and exploration of the solar system.
- Celestial mechanics.
- Detection methods and challenges.
- Telescopes and instrumentation. Ground vs space-based telescopes.
- Planet interiors.
- Surface processes.
- Planet atmospheres and atmospheric escape.
- The diversity of exoplanets, demographics and occurrence.
- Architecture of exoplanet systems.
- Introduction to planet formation theories.
- Habitability criteria.

Organisation

Lectures, exercises, discussions, project work and assignments.

Literature

Lecture notes complemented with articles and book chapters for instance from Planetary Sciences (de Pater and Lissauer) and The Exoplanet Handbook (Michael Perryman). The books can be downloaded from the Chalmers library website, and the articles are provided by the teachers.

Examination including compulsory elements

Written exam and compulsory assignments and project work.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers about disability study support.